Glio Core

PET · MRI · GLIOMAANALYSIS

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Methods Guide

Metabolic glioma segmentation from PET/MRI

Complete technical documentation version 1.0

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1. Introduction

GlioCore is an open-source Python tool for semi-automatic glioma segmentation from co-registered PET and MRI images in NIfTI format. The project stems from a specific methodological idea: to exploit the metabolic information of PET, where SUV and SUVR values distinguish hypometabolic necrosis from hypermetabolic active infiltration, information that MRI alone does not provide.

This guide documents in detail all implemented methods, the three operating modalities, the cluster-number validation suite, the white-matter atlas integration, the active-learning system and, above all, the exact way each metric is computed. The goal is full transparency and scientific reproducibility.

The three modalities of GlioCore

PET — SUVR + SUV + tumor mask (T1 optional). Metabolic segmentation.

MRI — T1 + T1ce + T2 + FLAIR + seg. For validation on the BraTS dataset.

PET+MRI — hybrid modality: uses all available channels without failing if some are missing.

2. Data architecture and feature set

The core of the multi-modal refactor is the clean separation between data loading and segmentation logic, mediated by an explicit data structure.

2.1 Automatic modality detection

The loader inspects the files present in the patient folder and determines the modality without user intervention: SUVR + SUV + mask activates PET; T1 + T1ce + T2 + FLAIR + seg activates MRI; PET data with additional MRI channels activates PET+MRI. No workaround such as reusing T1 as a fake SUVR is ever used: each modality loads its own real data.

2.2 The FeatureSet

All models receive an N×D feature matrix, where N is the number of tumor voxels and D the number of available channels. This allows both single-channel and multivariate segmentation.

Segmentation across all available channels

Unlike most clustering tools, which operate on a single intensity channel, GlioCore builds the feature matrix using all channels present simultaneously: up to four in MRI (T1, T1ce, T2, FLAIR) and up to six in PET+MRI (SUVR, SUV, T1, T1ce, T2, FLAIR). The system adapts automatically to the channels actually available, ignoring missing ones. This multivariate segmentation across the full channel space is a distinctive feature, active and exploited in the BraTS validation.

Field	Detail
PET	[SUVR, SUV] — primary feature SUVR
MRI	[T1, T1ce, T2, FLAIR] — primary feature T1ce
PET+MRI	all available channels, up to 6
Normalization	per-channel z-score (StandardScaler)
Cluster ordering	configurable on the primary feature, ascending or descending

Ordering clusters by the primary feature ensures a coherent biological meaning: cluster 1 corresponds to the lowest primary value (in PET, hypometabolism/necrosis), cluster k to the highest (hypermetabolism/infiltration). In MRI the default primary feature is T1ce, where enhancement appears hyperintense.

3. Segmentation models

GlioCore implements six models plus a baseline, all operating on the feature matrix of the current modality. Each declares explicitly which modalities it supports.

3.1 GMM — Gaussian Mixture Model

Field	Detail
Type	Parametric statistical, multivariate
k selection	minimized BIC or AIC
Library	sklearn.mixture.GaussianMixture
Reference	Zhao et al. (2021), Frontiers in Oncology, DOI: 10.3389/fonc.2021.679952

Models the feature distribution as a sum of k multivariate Gaussians estimated via Expectation-Maximization. Automatic k selection compares BIC/AIC values over a configurable range.

3.2 FCM — Fuzzy C-Means

Field	Detail
Type	Soft clustering, continuous membership [0,1]
k selection	fixed or automatic via Xie-Beni index
Library	scikit-fuzzy
Reference	Bezdek (1981), Pattern Recognition with Fuzzy Objective Function Algorithms

Each voxel receives a membership vector to all clusters rather than a hard label, capturing the ambiguity of transition zones between necrosis, edema and infiltration.

3.3 Level Set — Chan-Vese

Field	Detail
Type	Geometric contour evolution, intensity-based
Feature	uses only the primary channel (not multivariate)
Library	skimage.segmentation.morphological_chan_vese
Reference	Chan & Vese (2001), IEEE TIP, DOI: 10.1109/83.902291

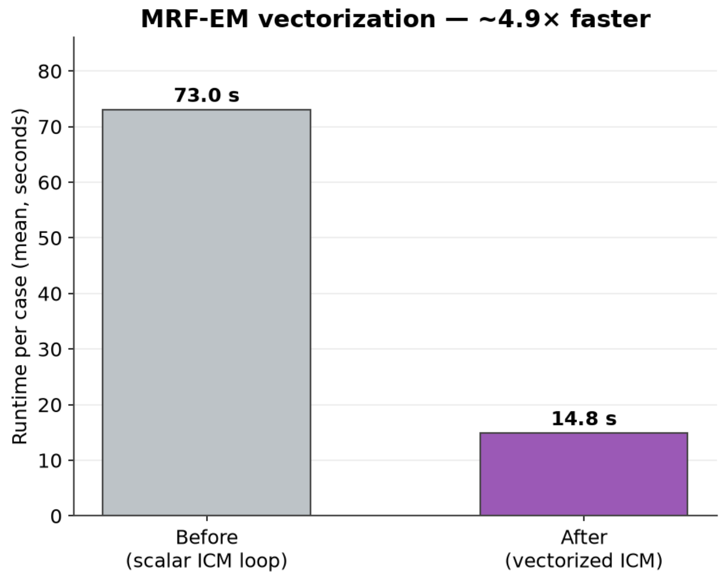
An active-contour model operating on a scalar field. By nature it uses a single channel (SUVR in PET, T1ce in MRI), a limitation declared explicitly. It produces spatially coherent contours, complementary to voxel-wise methods.

3.4 MRF-EM — Markov Random Field

Field	Detail
Type	Multivariate GMM + Ising spatial prior
β parameter	prior strength: 0 = pure GMM, 1.5 default
Algorithm	ICM (Iterated Conditional Modes)
Reference	Zhang et al. (2001), IEEE TMI, DOI: 10.1109/42.906424

Extends GMM by incorporating the biological prior that adjacent voxels tend to belong to the same tissue. It removes the salt-and-pepper noise of pure GMM, at the cost of higher runtime (voxel-by-voxel iterations).

Update - vectorized solver. The ICM step has since been vectorized: it produces identical segmentations but runs about 5x faster (~15 s per case instead of ~73 s), so MRF-EM is now competitive in runtime with the other methods.



MRF-EM runtime before and after vectorization of the ICM solver (mean over 271 BraTS cases); segmentation output is identical.

3.5 Adaptive DBSCAN

Field	Detail
Type	Density-based, non-parametric
Feature	modality channels + 3D spatial coordinates
Epsilon	estimated from the k-distance graph knee-point
Reference	Ester et al. (1996), KDD Proceedings

Requires no predefined k and finds clusters of arbitrary shape. Outlier voxels correspond to transition zones. It includes a fallback that widens ε if no clusters are found.

3.6 Hierarchical — GlioCore's original method

Original methodological contribution

This method is GlioCore's main contribution and achieves the highest structural agreement in the 271-case BraTS validation (ARI 0.52, equivalent to MRF-EM but about 3.4 times faster; see Validation Report).

Field	Detail
Type	Two-level hierarchical clustering
Level 1	separates inactive tissue (necrosis) from active tissue, weighting the primary feature
Level 2	subdivides only the active region into edema/enhancement
Parameters	primary_weight (default 1.5), n_level1 (default 2)
Runtime	~4.5 seconds per case
Conceptual reference	inspired by the BraTS hierarchy: Menze et al. (2015), IEEE TMI, DOI: 10.1109/TMI.2014.2377694

Instead of flat k-clustering, it proceeds hierarchically, mirroring the biological structure of the tumor. At the first level it separates metabolically inactive tissue (necrosis, low intensity) from active tissue. At the second level it subdivides only the active region into its components (edema and enhancement), leaving necrosis intact. The result is about three clusters that map naturally onto the three BraTS subregions.

This approach addresses the cluster-imbalance problem typical of flat clustering, where an under-represented subregion is fragmented or absorbed, and it is markedly faster than iterative methods such as MRF-EM.

3.7 Threshold — baseline

Field	Detail
Type	Quantile thresholding (not clustering)
Parameter	k = number of bands (default 3)
Feature	primary feature only (T1ce in MRI, SUVR in PET)
Role	minimum reference in the benchmark

A deliberately elementary method included as a lower reference. It divides voxels into k equally-sized bands by quantiles of the primary feature: it computes the thresholds at the quantiles (for k=3, at the 33rd and 66th percentile terciles) and assigns each voxel to the corresponding band via np.digitize. It performs no clustering and ignores the multivariate or spatial structure of the data.

Its purpose is to show how much value the clustering methods add: any sophisticated method should clearly surpass it. In the BraTS validation the baseline indeed obtains the lowest ARI (0.13), confirming that the structure captured by the clustering methods is not trivially reducible to intensity alone.

4. Cluster-number validation

The choice of k is critical and modality-dependent. On large datasets BIC alone tends to overestimate k because it rewards mathematical fit regardless of biological meaning. GlioCore implements a six-criterion suite with modality-adaptive weights.

4.1 The six criteria

Criterion	Direction	Meaning
Silhouette	maximize	cluster separation/cohesion
Calinski-Harabasz	maximize	inter/intra variance ratio
Davies-Bouldin	minimize	inter-cluster similarity
BIC / AIC	minimize	information criterion (reduced weight)
Elbow (inertia)	knee-point	knee of the inertia curve
Bootstrap stability	maximize	mean ARI over bootstrap samples

4.2 Modality-adaptive weights

The methodologically important point: not all criteria are equally reliable across modalities. In MRI, where features are four-dimensional, the Silhouette loses discriminative power (curse of dimensionality) and is therefore weighted less, in favor of Calinski-Harabasz and Davies-Bouldin which remain robust in high dimension.

Criterion	PET	MRI	PET+MRI
Silhouette	0.30	0.15	0.20
Calinski-Harabasz	0.25	0.30	0.28
Davies-Bouldin	0.25	0.30	0.28
BIC	0.10	0.15	0.14
Elbow	0.10	0.10	0.10

ARI is not included in k selection because it would require the ground truth, unavailable at this stage. It is used only in the BraTS benchmark where the ground truth exists. Reference for clustering validation: Rainio et al. (2025), arXiv:2502.07511.

5. How the metrics are computed

This section documents precisely the computation of each metric, because transparency on this point is essential for the scientific defensibility of the results.

5.1 Dice Similarity Coefficient

Measures the overlap between the predicted mask P and the ground truth G:

$$Dice = 2 \cdot |P \cap G| \,/\, (|P| + |G|)$$

It equals 1 for perfect overlap, 0 for no overlap. It is the standard metric in medical segmentation.

5.2 Jaccard Index

$$Jaccard = |P \cap G| \,/\, |P \cup G|$$

Related to Dice but stricter. It always holds that Jaccard ≤ Dice, a property verified in the quality checks.

5.3 Hausdorff Distance 95 (HD95)

The 95th percentile of the bidirectional surface distance between P and G, expressed in millimeters. Using the 95th percentile rather than the maximum makes it robust to isolated outliers. It measures contour accuracy, complementary to Dice which measures volume overlap.

5.4 Adjusted Rand Index (ARI)

The ARI measures the agreement between the partition produced by the model and the ground-truth classes, corrected for chance agreement. It is invariant to label permutation (it does not matter which number the model assigns to each cluster), and this makes it the most robust metric in the benchmark.

Why the ARI is the most reliable metric

The ARI compares the two partitions directly without using the ground truth to decide how to map clusters to regions. It therefore cannot be inflated by a-posteriori mapping choices. It is also permutation-invariant and chance-corrected.

Reference: Hubert & Arabie (1985), Journal of Classification, DOI: 10.1007/BF01908075.

5.5 The a-priori cluster-region mapping

A crucial methodological point. To compute the Dice of the BraTS subregions (Tumor Core and Enhancing Tumor) one must decide which model clusters correspond to which regions. GlioCore makes this choice a priori, based solely on the intensity order of the clusters, without ever looking at the ground truth.

- Enhancing Tumor (ET) = highest-intensity cluster, because enhancement is hyperintense on T1ce.
- Tumor Core (TC) = necrosis (lowest cluster) + enhancing (highest cluster), per the BraTS definition TC = label 1 + 3.

Why this matters

An approach that chose clusters by looking at the ground truth to maximize Dice would be metric overfitting: it would use the answer to build the answer, artificially inflating the numbers.

The a-priori intensity mapping is decided before seeing the ground truth, is biologically motivated, and is identical for all models. It yields lower but defensible numbers.

6. White-matter atlas

The JHU-ICBM atlas (48 tracts, MNI152 1mm space) makes it possible to quantify which white-matter tracts are involved by the lesion, analyzing separately the whole mask, the hypometabolic zone and the hypermetabolic zone.

1. Registration of the patient's T1 to MNI152 via ANTsPy (Affine ~1min or SyN ~5min).
2. Application of the same transformation to the masks (nearest-neighbor interpolation to preserve labels).
3. Voxel-wise overlap computation with the 48 JHU tracts, expressed as the percentage of each tract involved.
4. Automatic flagging of critical tracts (corticospinal tract, arcuate fasciculus, optic radiation).

References: Hua et al. (2008), NeuroImage, DOI: 10.1016/j.neuroimage.2007.07.053; Thiebaut de Schotten et al. (2018), GigaScience, DOI: 10.1093/gigascience/giy004.

7. Manual correction and active learning

Automatic segmentations can be corrected manually in the napari interface. Corrections are saved in a structured SQLite database with voxel-level metadata.

The BayesianRF model (Random Forest) trains incrementally on these corrections. On first run, in the absence of annotations, it starts from a synthetic bootstrap generated by GMM. Each correction session adds labeled samples, and the model improves progressively, adapting to the operator's annotation style. The model also computes uncertainty (Shannon entropy) for each voxel, identifying the regions that would most benefit from further annotation.

Reference: van der Voort et al. (2023), Neuro-Oncology, DOI: 10.1093/neuonc/noac166.

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